

Privacy-preserving federated learning for intrusion detection in smart factory IIoT networks, enabling real-time anomaly detection without exposing sensitive production data.

Manufacturing × Threat detection and response × Machine learning · OS001 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
48 is the world moving	74 can we play, can we win	MEDIUM 0 named in the evidence	€447m observed evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying privacy-preserving federated learning for intrusion detection in smart factory IIoT networks, enabling real-time anomaly detection without exposing sensitive production data. It targets manufacturing customers with distributed IIoT environments who need to detect cyber threats while complying with data privacy and sovereignty requirements. The timing is driven by the convergence of mature federated learning research and the urgent need for adaptive, privacy-preserving security in industrial settings.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€623m €64.5m – €4.9bn	€447m €46.2m – €3.5bn	€21.5m €2.2m – €167m	observed
Observed: tendered contract value, annualised	€50.5m €50.5m – €184m	€50.5m €50.5m – €184m	€2.4m €2.4m – €8.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€488k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (SA); the estimate covers the European footprint only.

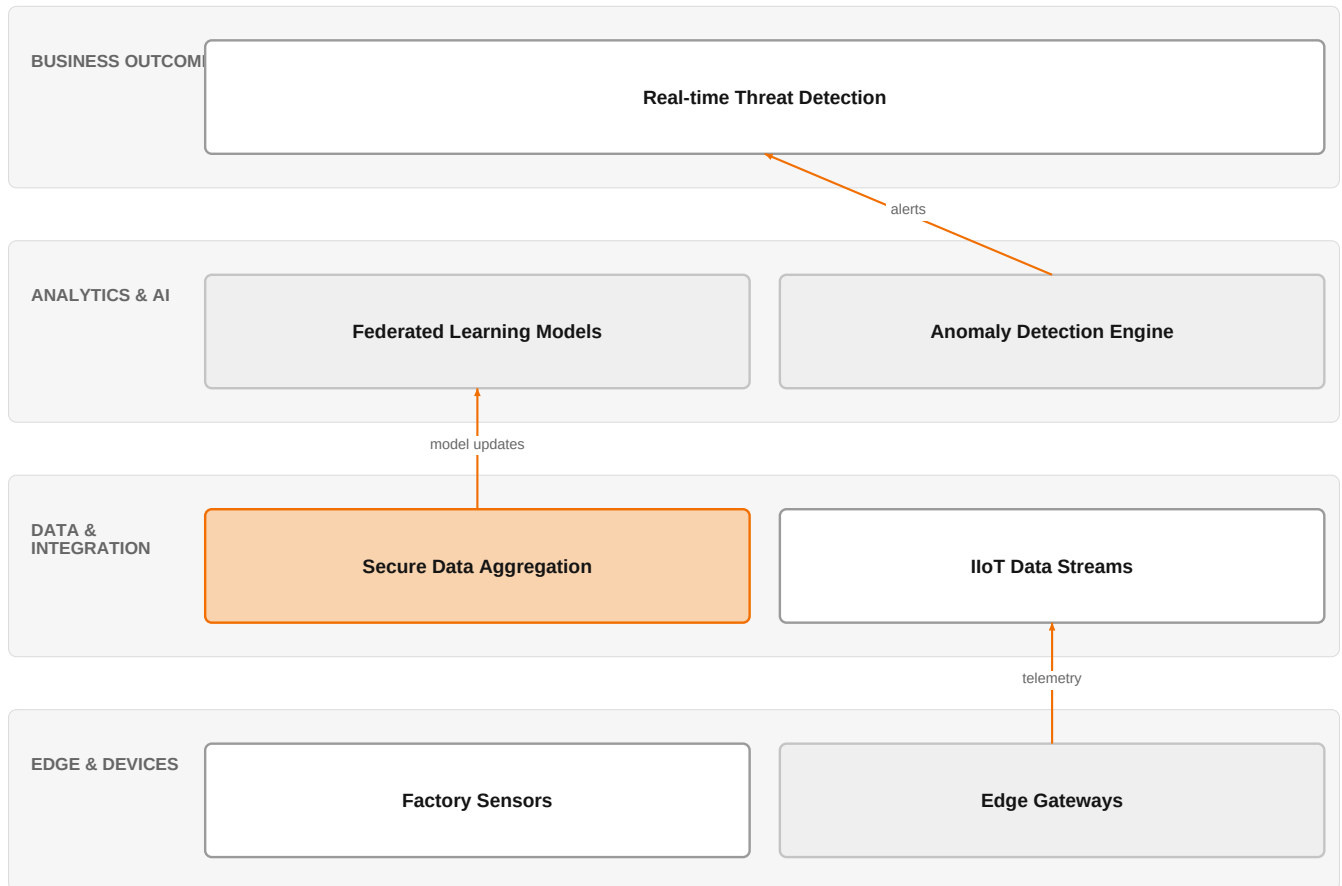
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€50.5m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 16 matching notices, matched on use case via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (SA); the estimate covers the European footprint only.

The solution, in one picture

Federated IIoT Anomaly Detection



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how federated learning enables privacy-preserving anomaly detection across distributed factory sensors, with Orange providing the AI/security expertise and secure aggregation via partners.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The market is shifting as federated learning frameworks for industrial anomaly detection have matured, with systematic reviews and multiple frameworks (e.g., SecuFL-IIoT, ROCHE) demonstrating adaptive, privacy-preserving capabilities. Traditional centralized intrusion detection systems are failing against zero-day attacks and lack scalability for high-volume IIoT traffic, while privacy concerns prevent sharing sensitive production data. Research shows a clear trend toward decentralized, collaborative models that preserve data privacy while enabling real-time threat detection.

Sources: SIG-C6ED90595B8C openalex.org 2026-04-18; SIG-F633044AF87A openalex.org 2026-01-29; SIG-CF12E7BF5186 openalex.org 2025-09-22; SIG-8EC0D797B8D2 openalex.org 2025-10-27; SIG-CEE282CB905A openalex.org 2025-09-08

Who buys, and why now

The buyer is typically the Chief Information Security Officer (CISO) or Head of Industrial Cybersecurity at a manufacturing firm, while the operational pain is felt by plant managers and OT security teams who face increasing cyber threats and the challenge of detecting anomalies without disrupting production or exposing proprietary process data. The trigger is often a security incident, an audit finding, or a regulatory pressure to improve threat detection while maintaining data sovereignty.

Why it is hot now — every claim bound to a dated source

- Federated learning frameworks enable adaptive and privacy-preserving anomaly detection in smart industrial networks. [SIG-F633044AF87A]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution leveraging its Cybersecurity experts and AI/Data/Cloud experts to design and deploy a federated learning-based intrusion detection system. The engagement would start with an assessment of the customer's IIoT architecture and data flows, then build a proof-of-concept using Orange's managed services and possibly partner cloud infrastructure (e.g., AWS or Google Cloud) for secure aggregation. The solution would be certified against ISO 27001 and TISAX where relevant, and could integrate with Orange Cyberdefense managed services for ongoing monitoring and response.

Why Orange can win

Orange can win by combining its deep cybersecurity expertise with its ability to deliver sovereign, certified solutions (ISO 27001, TISAX) that address data privacy concerns. The presence of Orange Group researchers and defense/security experts adds credibility in advanced AI and security. However, the assets are thin in terms of a specific federated learning product, so Orange would need to partner with cloud providers or leverage its managed services to assemble a credible offering.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NICE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include hyperscalers AWS and Google Cloud, which offer machine learning platforms but may lack deep industrial OT security expertise. Atos/Eviden competes in both machine learning and cybersecurity, potentially offering

integrated solutions. NiCE, a customer-experience platform, is also an Orange partner and sells machine learning. Data and AI platforms like Databricks, Dataiku, and Palantir offer advanced analytics but may not have industrial-specific security focus. Accenture is active in manufacturing and could bring systems integration capabilities. The customer will compare Orange's ability to combine security, sovereignty, and industrial domain knowledge against these alternatives.

Sources: SIG-F633044AF87A openalex.org 2026-01-29; SIG-CF12E7BF5186 openalex.org 2025-09-22

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning; competes in Cybersecurity	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning	structural	—
Accenture	Global systems integrator	active in Manufacturing	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Do you currently have any intrusion detection systems in place for your IIoT networks, and if so, how do they handle data privacy?
- 2 Have you experienced any security incidents or near-misses that were difficult to detect due to data silos or privacy constraints?
- 3 What are your main concerns about sharing production data with a central security analytics platform?
- 4 Are you exploring federated learning or other privacy-preserving AI techniques for your industrial security?
- 5 Do you have a distributed IIoT environment across multiple sites that would benefit from collaborative threat detection?

Objections you will meet

They will say	You can answer
Federated learning is too immature for production use in our factory.	You're right that it's still emerging, but the research is converging on practical frameworks. We can start with a proof-of-concept on a limited set of sensors to validate performance and then scale gradually.
We already have a security solution from a hyperscaler.	Hyperscalers offer strong ML platforms, but they may not have deep OT security expertise or the sovereign, certified infrastructure that Orange can provide. We can complement your existing tools with a federated layer that respects data privacy.
We're concerned about the cost and complexity of implementing this.	That's a valid concern. We can design a phased approach, starting with a small pilot to demonstrate value, and we can leverage our managed services to reduce operational burden.

Risks and unknowns

The main risk is that federated learning for IIoT intrusion detection is still largely in research phase, with limited production deployments. There are unknowns about the practical performance in real factory environments, the overhead of secure aggregation, and the ability to integrate with existing OT systems. Additionally, the competitive landscape is crowded, and Orange's lack of a proprietary federated learning platform could be a disadvantage unless partnerships are leveraged effectively.

Next action, by role (FR-17)

Role	Do this next
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Presales / Proposal	Prepare a bid brief for a smart factory client that positions Orange Drone Guardian and Orange Cyberdefense managed services alongside ISO 27001 and TISAX certifications, leveraging the Colgate-Palmolive and Saint-Gobain Glass references to illustrate privacy-preserving anomaly detection.
Sales	In a conversation with a manufacturing CIO, say: 'We're exploring federated learning to detect anomalies in your IIoT network without exposing production data—could we discuss how Orange Cyberdefense managed services could support a pilot?'
Strategist / Innovator	Commission a deep-dive brief on federated learning for IIoT intrusion detection, mapping Orange's AI/Data/Cloud and cybersecurity experts against the capabilities of AWS, Google Cloud, and NVIDIA, and validating the approach with Orange Group researchers.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D2558A0D8EDD	2026-08-01	Computer Communications	1	Unsupervised CoAP-based anomaly detection in private 5G core network for Industrial-IoT
SIG-C6ED90595BBC	2026-04-18	openalex.org	1	Towards collaborative and privacy-preserving industrial time-series anomaly detection with fede...
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-1C479A2B44CD	2025-11-01	openalex.org	1	CoperFed: A covert personalized federated learning framework for Industrial Control Systems int...
SIG-904D344AF0C0	2025-10-28	openalex.org	1	Federated learning based on two-stage knowledge distillation for intrusion detection in industr...
SIG-C8282154F431	2025-10-28	openalex.org	1	Lightweight Signal Processing and Edge AI for Real-Time Anomaly Detection in IoT Sensor Networks
SIG-8EC0D797B8D2	2025-10-27	openalex.org	1	AI-Powered Security for IoT Ecosystems: A Hybrid Deep Learning Approach to Anomaly Detection
SIG-A49A291BF70B	2025-10-01	openalex.org	1	PureChain-enhanced federated learning for dynamic fault tolerance and attack detection in distr...
SIG-CF12E7BF5186	2025-09-22	openalex.org	1	ROCHE: A Robust and End-to-End Privacy-Preserving Federated Learning Framework for Intrusion De...
SIG-B75BEF75986F	2025-09-08	openalex.org	1	Trustworthy Adaptive AI for Real-Time Intrusion Detection in Industrial IoT Security
SIG-CEE282CB905A	2025-09-08	openalex.org	1	Survey of Federated Learning for Cyber Threat Intelligence in Industrial IoT: Techniques, Appli...
SIG-8FD8EE65E437	2025-08-19	openalex.org	1	Differential privacy-enabled federated learning for secure neural synchronization in protecting...
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.